Linear Algebra Lesson 3: Vector Spaces

Grace Xu

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Finally the word vector appears! As I wrote in our earlier letters, since we are going about linear algebra in a proof-based abstract manner, vector spaces are dealt with quite interestingly - by which I mean abstractly. Thus I think your studies in your own linear algebra textbook will be a useful and practical supplement here. Vectors are extremely useful in data analytics and machine learning. However, in our correspondence, we shall more so learn the abstract foundations underlying vectors.

I hope you are prepared for an onslaught of definitions and notations. When I first learned these, I once again thought it was fascinating how such simple concepts like addition and multiplication need to be defined, since we learn them in elementary school. It was cool relearning these concepts and seeing how in elementary school, we are just taught the intuition, but now in higher-level math, we can define them more rigorously. It was also fascinating to realize how much one learned from just intuition as a little kid! Lastly, I will preface that from now onwards, we will largely be dealing with vectors rather than plain old numbers (which will from now on be called scalars).

1. **Notation: Vector**
   Vectors are denoted with an arrow above the vector name, e.g. $\vec{v}$. Geometrically a vector is like an arrow. Algebraically vectors are an ordered list of coordinates, and sometimes there are pointy brackets $<>$ to denote that it is a vector specifically, though I will often just use the regular ( ). So in the xy-plane which is 2D, an example vector is $\vec{v} =< 1, -2 >$. In 3D, an example vector is $\vec{u} =< -2, 3, 1 >$.

   (a) Definition: A **vector** is a geometric object with both magnitude (length) and direction.

2. **Notation: Fields**

   (a) **Notation:** $\mathbb{F}$ denotes a field of scalars. I like to conceptualize this as the field of regular numbers you are used to (this is not rigorous; just my personal thought process).

   From now on, when you see $\mathbb{F}$, you can assume it to be either $\mathbb{R}$ (the real numbers) or $\mathbb{C}$ (the complex numbers). There are other fields like we discussed last time; but from now on $\mathbb{F}$ is usually referencing $\mathbb{R}$ or $\mathbb{C}$ (because these are the most relevant fields).

   (b) **Notation:** $\mathbb{F}^n$ is used to denote higher dimensions (and is a vector space as we will later learn). I think of it this way: $\mathbb{F}$ is 1-dimensional (like a number line), $\mathbb{F}^2$ is 2D (like an xy-graph), $\mathbb{F}^3$ is 3D, and so on.

   (c) **Example:** $< 2 > \in \mathbb{F}$ (visualized as a dot on a number line),
   $< 3, 1 > \in \mathbb{F}^2$ (visualized as an arrow in a flat plane),
   $< 5, -1, 3 > \in \mathbb{F}^3$ (visualized as an arrow in a 3D space)

3. **Definition: $\mathbb{F}^n$, Coordinates**

   (a) $\mathbb{F}^n$ is the set of all lists of length n of elements of $\mathbb{F}$:

   $$\mathbb{F}^n = \{(x_1, x_2, ..., x_n) | x_1, x_2, ..., x_n \in \mathbb{F}\}$$

   (b) For any $(x_1, x_2, ..., x_n) \in \mathbb{F}^n$, the numbers $x_1, x_2, ..., x_n$ are referred to as **coordinates**. x_2 is referred to as the 2nd coordinate, x_n is referred to as the n-th coordinate, etc.

4. **Definition: Addition, Scalar multiplication in $\mathbb{F}^n$**
   These definitions may seem simple, but it is important to define them clearly so that we can do more complicated proofs later.

   (a) **Addition in** $\mathbb{F}^n$: For $(x_1, ..., x_n), (y_1, ..., y_n) \in \mathbb{F}^n$, we define

   $$(x_1, ..., x_n) + (y_1, ..., y_n) = (x_1 + y_1, ..., x_n + y_n)$$

   (b) **Example**: $(2, 3) + (1, 4) = (2 + 1, 3 + 4) = (3, 7)$
   (which one could also write as $< 2, 3 > + < 1, 4 > = < 2+1, 3+4 > = < 3, 7 >$).

   (c) **Scalar multiplication in** $\mathbb{F}^n$: For $(x_1, ..., x_n) \in \mathbb{F}^n$, $\lambda \in \mathbb{F}$ we define

   $$\lambda(x_1, ..., x_n) = (\lambda x_1, ..., \lambda x_n)$$

   (d) **Example**: $5(2, 3, 4) = (5 \cdot 2, 5 \cdot 3, 5 \cdot 4) = (10, 15, 20)$.

5. **Proposition**: Commutativity in $\mathbb{F}^n$

   $$\vec{x} + \vec{y} = \vec{y} + \vec{x} \text{ for all } \vec{x} + \vec{y} \in \mathbb{F}^n$$

   (a) **Proof:** One of our first proofs!
   Let's write $\vec{x} = (x_1, ..., x_n)$ and $\vec{y} = (y_1, ..., y_n)$.
   Then $\vec{x} + \vec{y} = (x_1, ..., x_n) + (y_1, ..., y_n)$ by definition of $\vec{x}, \vec{y}$
   $= (x_1 + y_1, ..., x_n + y_n)$ by definition of addition in $\mathbb{F}^n$
   $= (y_1 + x_1, ..., y_n + x_n)$ by commutativity of addition in $\mathbb{F}$ since the elements of $\vec{x}, \vec{y}$ are scalars in $\mathbb{F}$

$= (y_1, ..., y_n) + (x_1, ..., x_n)$ by definition of addition in $\mathbb{F}^n$
$=\vec{y} + \vec{x}$ by definition of $\vec{x}, \vec{y}$.
QED.

(b) Here I wrote out in English the reasoning behind every line of the proof. This is good practice to make sure one understands why each step of the proof is possible. As you can see, in this proof we cited many definitions. As we learn more, we will also begin to cite propositions, theorems, or even axioms.

6. **Definitions: Addition, Scalar Multiplication in Vector Spaces**

   (a) An **addition** on a set V is a function that assigns an element $\vec{u}+\vec{v} \in V$ to each pair of elements $\vec{u}, \vec{v} \in V$

   (b) A **scalar multiplication** between a set V and a field $\mathbb{F}$ is a function that assigns an element $\lambda \cdot \vec{v}$ for each pair $\vec{u} \in V$, $\lambda \in \mathbb{F}$

   (c) **Note:** You will notice that V is commonly used to denote a vector space. This is just etiquette, not any formal definition. Similarly, λ is commonly used to denote a scalar, but sometime we just use regular letter variables such as a, b, x, y etc.

7. **Definition: Vector Space over a field $\mathbb{F}$**
   This is a big definition - and super important!

   A **vector space** over a field $\mathbb{F}$ is a set V with an addition function and a scalar multiplication function with $\mathbb{F}$, and which also satisfies the following properties:

   (a) **Commutative Property:**
      i. $\vec{x} + \vec{y} = \vec{y} + \vec{x}$ for all $\vec{x}, \vec{y} \in V$

   (b) **Associative Property:**
      i. $(\vec{x} + \vec{y}) + \vec{z} = \vec{x} + (\vec{y} + \vec{z})$ for all $\vec{x}, \vec{y}, \vec{z} \in V$
      ii. $(ab)\vec{v} = a(b\vec{v})$ for all $a, b \in \mathbb{F}$ and $\vec{v} \in V$

   (c) **Distributive Property:**
      i. $a(\vec{x} + \vec{y}) = a\vec{x} + a\vec{y}$ for all $a \in \mathbb{F}$ and $\vec{x}, \vec{y} \in V$
      ii. $(a + b)\vec{x} = a\vec{x} + b\vec{x}$ for all $a, b \in \mathbb{F}$ and $\vec{v} \in V$

   (d) **Identities:**
      There exists the distinct elements $\vec{0}, \vec{1} \in V$, also known as identities, such that
      i. Additive Identity ($\vec{0}$): $\vec{x} + \vec{0} = \vec{x}$ for all $\vec{x} \in V$
      ii. Multiplicative Identity ($\vec{1}$): $1\vec{x} = \vec{x}$ for all $\vec{x} \in V$

   (e) **Additive Inverse:**
      i. For all $\vec{x} \in V$, there exists a unique $\vec{y} \in V$ such that $\vec{x} + \vec{y} = \vec{0}$

8. **Some asides on vector spaces**
   First I just wanted to note than any terms without arrows over them are scalars, while those with arrows are vectors.

   Although there are now a lot of vectors in this definition, you might notice that it still looks quite familiar. It looks an awful lot like the definition of a field!

   There is a key difference: in the definition of a vector space, we do not have a multiplicative inverse.

   This is because a vector space only has addition and **scalar** multiplication, whereas a field has addition and (regular old) multiplication. What does this mean?

   Well, in a field, you can multiply different elements (e.g. numbers) of the field together.

   Within the definition of a vector space, we only really discuss scalar multiplication, which is multiplying a scalar (not an element of the vector space; rather it is an element of a field) with a vector (an element of the vector space).

   Ultimately, I think of it this way: a field contains just regular numbers, and a vector space contains, well, vectors.

   Earlier, we discussed $\mathbb{F}^n$. So actually, fun fact, $\mathbb{F}^n$ is a vector space over the field $\mathbb{F}$ given n is a natural number. (E.g. $\mathbb{R}^n$ is a specific type of $\mathbb{F}^n$, over the field $\mathbb{R}$.)

   To prove that $\mathbb{F}^n$ is a vector space, we would need to prove it satisfies all the properties in the definition of a vector space, including commutativity, associativity, distributivity, identities, and additive inverse. We actually already proved its commutativity property in the proof above (5(a)). The proofs of the remaining properties are either shown below or left as exercises.

9. **Definition: $\vec{0}$, the zero vector**
   For $\mathbb{F}^n$, we define $\vec{0} = (0, ..., 0)$ such that there are n zeros.

   (a) Examples:
   In $\mathbb{F}^5$, $\vec{0} = (0, 0, 0, 0, 0)$
   In $\mathbb{F}^2$, $\vec{0} = (0, 0)$
   Note that $(0, 0, 0, 0, 0) \neq (0, 0)$. These are different objects!

10. **Proposition: $\vec{0}$ is additive identity in $\mathbb{F}^n$**
    See 7)d)i).

$$\vec{x} + \vec{0} = \vec{x} \; \forall \; \vec{x} \in \mathbb{F}^n$$

Proof: Let $\vec{x} = (x_1, ..., x_n)$. Then,
$\vec{x} + \vec{0} = (x_1, ..., x_n) + (0, ..., 0)$ by definition of $\vec{x}, \vec{0}$
$=(x_1 + 0, ..., x_n + 0)$ by definition of addition in $\mathbb{F}^n$
$=(x_1, ..., x_n)$ by additive identity in $\mathbb{F}$, since the scalar 0 is additive identity in $\mathbb{F}$
$= \vec{x}$ by definition of $\vec{x}$. QED.

11. **Proposition: Additive inverses in** $\mathbb{F}^n$
See 7)e).

$$\forall \vec{x} \in \mathbb{F}^n \;\; \exists\, \vec{y} \in \mathbb{F}^n \text{ such that } \vec{x} + \vec{y} = \vec{0}$$

Proof: Let $\vec{x} = (x_1, ..., x_n)$. Take $\vec{y} = (-x_1, ..., -x_n)$. This is possible because in our last lesson, we saw that additive inverses do exist in $\mathbb{F}$. Then,
$\vec{x} + \vec{y} = (x_1, ..., x_n) + (-x_1, ..., -x_n)$ by definition of $\vec{x}, \vec{y}$
$=(x_1 + (-x_1), ..., x_n + (-x_n)$ by definition of addition in $\mathbb{F}^n$
$=(0, ..., 0)$ by additive identity in $\mathbb{F}$
$= \vec{0}$ by definition of $\vec{0}$. QED.

12. **Proposition: A vector space has a *unique* additive identity,** $\vec{0}$

Proof: Suppose $\vec{0}, \vec{z}$ are both additive identities. Our goal is to show $\vec{z} = \vec{0}$, i.e., $\vec{0}$ is the only additive identity. Then,
$\vec{z} = \vec{z} + \vec{0}$ because $\vec{0}$ is an additive identity
$= \vec{0} + \vec{z}$ by commutative property
$= \vec{0}$ as we defined $\vec{z}$ to be an additive identity.
Thus, we have shown $\vec{z} = \vec{0}$, i.e., the additive identity $\vec{0}$ is unique. QED. QED.

13. **Proposition: Every element in a vector space has a *unique* additive inverse,** $\vec{-u} \; \forall \, \vec{u} \in V$

Proof: Suppose $\vec{u} \in V$; $\;\vec{v}, \vec{v}' \in V$ where $\vec{v}, \vec{v}'$ are both additive inverses, i.e. $\vec{u} + \vec{v} = \vec{0}$ and $\vec{u} + \vec{v}' = \vec{0}$. Our goal is to show $\vec{v} = \vec{v}'$, i.e., the additive inverse is unique. Then,
$\vec{v} = \vec{v} + \vec{0}$ because $\vec{0}$ is an additive identity
$= \vec{v} + (\vec{u} + \vec{v}')$ since we defined $(\vec{u} + \vec{v}' = \vec{0})$
$= (\vec{v} + \vec{u}) + \vec{v}'$ by associative property
$= (\vec{u} + \vec{v}) + \vec{v}'$ by commutative property
$= (\vec{0}) + \vec{v}'$ since we defined $(\vec{u} + \vec{v} = \vec{0})$
$= \vec{v}' + \vec{0}$ by commutative property
$= \vec{v}'$ by additive identity.
Thus, we have shown $\vec{v} = \vec{v}'$, i.e., the additive inverse is unique. QED.

14. **Notation:** $\vec{u} + (\vec{-v}) = \vec{u} - \vec{v}$

15. **Exercises**
Below are some exercises regarding vector spaces.

(a) Prove that the distributivity property of vector spaces (See 7)c) holds for $\mathbb{F}^n$, i.e., show

$$\lambda(\vec{x} + \vec{y}) = \lambda\vec{x} + \lambda\vec{y} \; \forall \; \lambda \in \mathbb{F}, \; \vec{x}, \vec{y} \in \mathbb{F}^n$$

(b) Prove that the scalar multiplicative identity property of vector spaces (See 7)d)ii) holds for $\mathbb{F}^n$, i.e., show

$$1\vec{x} = \vec{x} \quad \forall \; \vec{x} \in \mathbb{F}^n \text{ where } 1 \in \mathbb{F}$$

(c) Find $\vec{x} \in \mathbb{R}^4$ such that $(4, -3, 1, 7) + 2\vec{x} = (5, 9, -6, 8)$

(d) Prove $-(-\vec{v}) = \vec{v} \; \forall \; \vec{v} \in V$

(e) Given $a \in \mathbb{F}, \; \vec{v} \in V,$ and $a\vec{v} = \vec{0},$ prove $a = 0$ or $\vec{v} = \vec{0}$. (Hint: Consider 2 cases: $a = 0$ or $a \neq 0$,)